Final Project — Advanced Simulation Study

Modules M01–M12 | Chapters 1–14 | Weight 25% | Difficulty ⋆⋆⋆

Modules: M01–M12 Chapters: 1–14 Weight: 25% Due: Finals Week

Overview

The final project is an individual (or pairs with instructor approval) advanced simulation study. It builds on your midterm system or introduces a new, more complex system. Every submission must include the core simulation study components **plus one advanced extension** from the options below.

1 Advanced Extension Options

Choose **exactly one** extension.

Option A — Simulation Optimization (Module M11)

Apply a systematic search over a design space to find the best policy or configuration.

Required additions to the core study:

- Decision variable definition and feasible set
- Search strategy: grid search with refinement, RSM/surrogate, or approved gradient-free method
- Replication budget justification (pilot vs. final validation)
- Confidence interval for the recommended design
- Comparison against the midterm baseline

Full specification: `assignments/final_project_simopt_option.md`

Option B — Reinforcement Learning (Module M12)

Wrap the simulation as a Gymnasium environment and train an RL agent to control it.

Required additions to the core study:

- MDP formulation: state, action, reward, episode structure
- Gym-compatible wrapper (built on `simdes.envs.base_env.SimPyEnv`)
- Training curve (episode return vs. training steps)

- Policy evaluation: RL agent vs. at least one hand-crafted heuristic
- Discussion of reward shaping choices and their effect on agent behavior

Full specification: `assignments/final_project_rl_option.md`

2 Core Components (All Submissions)

2.1 1. System Design (15 pts)

- More complex than the midterm: at least two resource types, non-trivial routing or priority logic, or time-varying demand
- Complete conceptual model with updated diagram and assumptions (≥ 6)
- ≥ 3 performance measures; all with units answering the stated research question

2.2 2. Implementation and V&V (20 pts)

- Modular, class-based SimPy code; parameters in `dataclass`; seeds as arguments
- Verification: at least **two** extreme-condition tests and one conservation check
- Validation: face validity statement with domain expert reference or published benchmark
- Same seed $\Rightarrow$ same output; clean execution from repo root

2.3 3. Output Analysis (15 pts)

- Warm-up analysis with method and ℓ^* stated in simulation-time units
- ≥ 30 independent replications; 95% CIs for all primary metrics
- Replication summary table: metric, mean, half-width, n

2.4 4. Advanced Extension (35 pts)

See Option A or Option B above. See also `rubrics/final_project_rubric.pdf`.

2.5 5. Report and Presentation (15 pts)

- Written report: ≤ 15 pages (excluding figures, tables, code appendix)
- 10-minute in-class presentation + 5-minute Q&A (last week of term)
- Presentation must convey: system, method, key result, and one limitation

3 Report Structure

Section	Content
1	Executive Summary ($\leq$ 200 words)
2	System Description and Motivation
3	Conceptual Model
4	Input Modeling
5	Implementation and V&V
6	Baseline Output Analysis
7	Advanced Extension Option A: optimization problem, method, results Option B: MDP design, training, policy evaluation
8	Discussion and Recommendation
9	Limitations and Future Work
–	References
–	Appendix: Code listings (not counted toward page limit)

4 Deliverables

Item	Format	Notes
Written report	PDF, $\leq$ 15 pages	Submitted via GitHub Classroom
Code repository	GitHub Classroom	Must run after `pip install -e simdes/`
Presentation slides	PDF	Submitted day of in-class presentation

Include a `Makefile` or `README` with a single command to reproduce all key figures.

Grading Summary

Component	Points
System design and conceptual model	15
Implementation and V&V	20
Baseline output analysis	15
Advanced extension (A or B)	35
Report and presentation	15
Total	**100**

Timeline Checkpoints

Date	Checkpoint	Action Required
Week 13	Topic confirmation	1-paragraph system description to instructor
Week 14	Model checkpoint	Share running SimPy model (output analysis not required yet)
Week 15	Presentation	In-class; slides submitted day-of
Finals	Final report	GitHub Classroom submission

Pairing Policy

Pairs are allowed with instructor approval. Pair submissions must demonstrate substantially greater scope than solo submissions:

- At least three scenarios (vs. two for solo)
- Both advanced extensions explored, even if only one is the primary focus
- Joint presentation with each partner speaking for ≥ 5 minutes

Points are shared equally unless one partner files a contribution statement indicating a different split (approved by instructor at the Week 14 checkpoint).

Full rubric: `rubrics/final_project_rubric.pdf` | *Solutions and exemplar reports: the instructor package (available to instructors on request).*